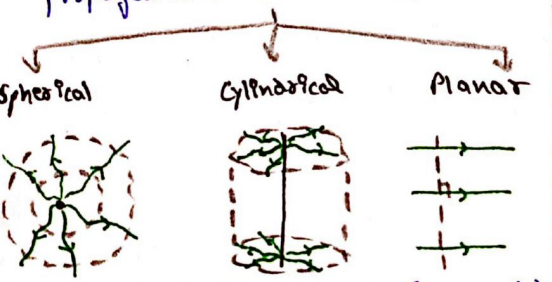
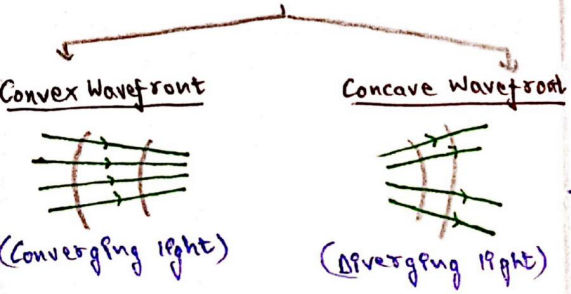
Wave Optics

Wavefront :

→ When light propogates, the cross section where all particles oscillates in same phase is **Wavefront**

→ Wavefront is always Normal to propogation direction.



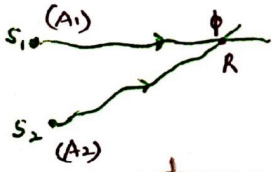


Interference of light Wave :

Condition ⇒ Light sources must be coherent i.e

↳ Same frequency (f & ω)

↳ Const phase difference.



Amplitude	Intensity
$R = A_1^2 + A_2^2 + 2A_1A_2\cos\phi$	$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$
If $A_1 = A_2 = A$	If $I_1 = I_2 = I_0$
$R = 2A\cos\phi/2$	$I_R = 4I_0\cos^2(\phi/2)$
Also,	Also,
$R_{max} = (A_1 + A_2)^2$	$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$
$R_{min} = (A_1 - A_2)^2$	$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

→ Interference can be of two types as following :

(I) Constructive Interference :→

Cond^n ⇒ (i) $\Delta\phi = 2n\pi$

(ii) $\Delta x = n\lambda$

→ $R = A_1 + A_2 = 2A$

$I_R = (\sqrt{I_1} + \sqrt{I_2})^2 = 4I_0$

(II) Destructive Interference :→

Cond^n ⇒ (i) $\Delta\phi = (2n+1)\pi$

(ii) $\Delta x = (2n+1)\frac{\lambda}{2}$

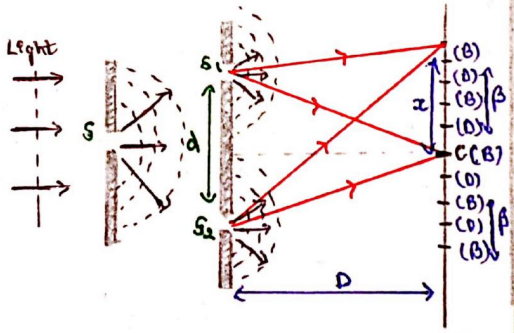
→ $R = A_1 - A_2 = 0$

$I_R = (\sqrt{I_1} - \sqrt{I_2})^2 = 0$

YDSE :

→ Experiment to practically demonstrate interference of light.

→ At standard conditions, path diff. at centre is zero & Bright fringe is obtained.



→ S_1 & S_2 are coherent sources, because both are recieving light from S, symmetrically

→ Path diff b/w light waves at x is given by

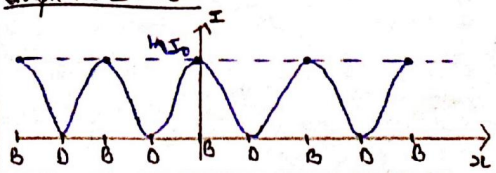
$$\Delta = \frac{dx}{D}$$

$$\phi = \frac{2\pi}{\lambda}\cdot\Delta \Rightarrow \frac{2\pi}{\lambda}\cdot\frac{dx}{D}$$

So, $I(x) = 4I_0\cos^2(\phi/2)$

$$\Rightarrow I(x) = 4I_0\cos^2\left(\frac{\pi dx}{\lambda D}\right)$$

Graph b/w I & x



Distance of nth bright fringe from C

$$x_{bright} = \frac{n\lambda D}{d}$$

Distance of nth dark fringe from C

$$x_{dark} = \frac{(2n-1)\lambda D}{2}$$

Fringe Width (β) ⇒

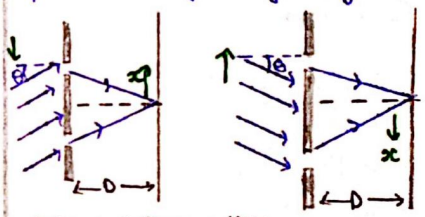
It is distance b/w two consecutive dark or bright fringes.

$$\beta = \frac{\lambda D}{d} \qquad \beta_\theta = \frac{\lambda}{d}$$

(Angular Fringe Width)

Changing Direction of Incident light in YDSE :

→ Iss direction mein light ko bend karenge, uske opp dir^n mein fringe pattern shift ho jaayega.



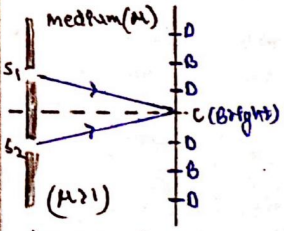
Shift of fringe pattern

$$x = D\sin\theta$$

Effect of Submerging YDSE set up in a Transparent medium :

→ No effect on Intensity of light.

→ Just, the fringe width decreases & number of fringes ↑ses.



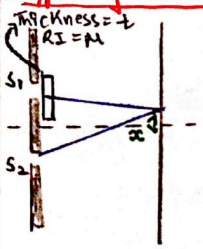
$(\lambda' = \lambda/\mu)$

If $\beta = \frac{\lambda D}{d}$

$$\beta' = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

$(\beta' < \beta)$

Effect of Insertion of thin Film in YDSE :



Shift of fringe pattern

Path diff due to film = Path diff. at x

$$t(\mu - 1) = \frac{dx}{D}$$

$$\Rightarrow x = \frac{Dt(\mu-1)}{d}$$

→ Due to film at S_1, pattern shifts up↑

→ Due to film at S_2, pattern shifts down↓

→ If films (μ_1 & μ_2) are present at both slits, of equal thickness, then

$$x = \frac{Dt(\mu_1 - \mu_2)}{d}$$

Use of White Light in YDSE :

For a white light

$\lambda_{violet} \simeq 4000\,Å$

$\lambda_{Red} \simeq 8000\,Å$

For violet Ray

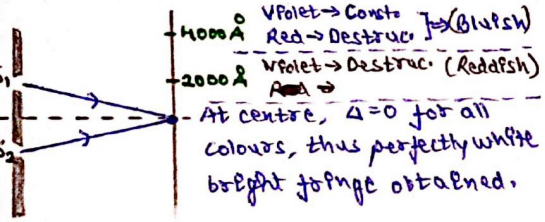
Const. Interference ⇒ 0, 4000, 8000, 12000 ---

Destr. Interference ⇒ 2000, ~~8000~~ 6000, 10000 ---

For Red Ray

Const. Interference ⇒ 0, 8000, 16000, 24000 ---

Destr. Interference ⇒ 4000, 12000, 20000 ---

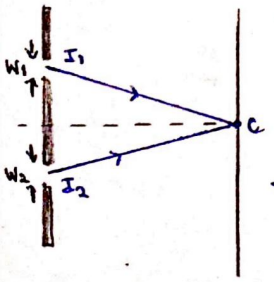


→ As we move little away, we observe 'Reddish Bright Fringe' (Destructive interference of violet)

→ If move further more away, we observe 'Bluish Bright Fringe' (Destructive interference of Red)

Effect of changing slit width of YDSE :

→ Intensity of light wave (I) ∝ slit width (W)



→ Due to different slit widths, we don't observe contrasting fringe pattern.

→ Because I_{max} & I_{min} are some finite values & not $4I_0$ & 0, here.

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

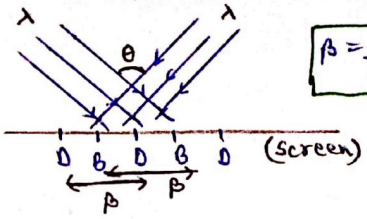
$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Constrast Ratio

$$r_c = \frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2$$

Interference of two Converging Coherent Parallel beams :

→ If two coherent beams interfere, they form YDSE like fringe pattern on screen.

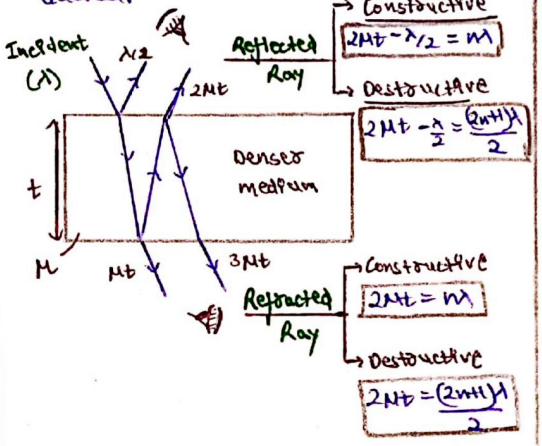


$$\beta = \frac{\lambda}{\theta}$$

Interference in Thin Films :

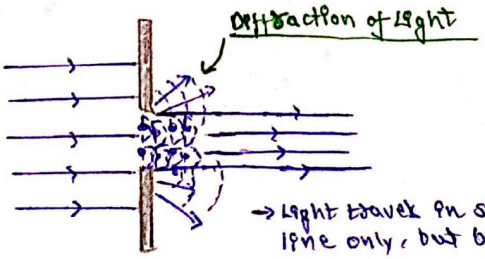
→ We consider only near normal incidence.

→ When reflection is from denser medium, λ/2 path difference is added.



Diffraction of Light :

→ Diffraction is bending of light at the edges of obstacles placed in path of the beam.



→ Light travels in st. line only, but by Hygens theory, we know that each point behaves like a secondary source of light.

→ So, with distance, the terminal secondary source flairs out from rectilinear path.

→ And besides straight light beam, some light beam is also observed outside the rectilinear path.

Type of Diffraction :

→ Jitni dur screen hogi, flared out light ka utna kam intensity screen pe ~~kam~~ observe hogi.

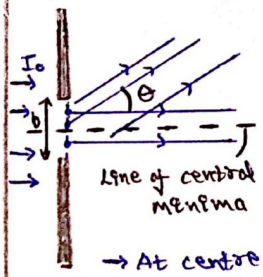
Intensity of flared out light ∝ D

Fresnel Diffraction — For a finite small distance observation

Fraunhoffer Diffract — For very far or infinite distance observation.

Diffraction by a Single Slit :

→ Specific case of Fraunhoffer Diffraction.



$$I_\theta = I_0 \frac{\sin^2\beta}{\beta}$$

$$\beta = \frac{\pi b \sin\theta}{\lambda}$$

→ At centre, most bright fringe is obtained, because

$\theta = 0° \Rightarrow \beta = 0$; $I_\theta = I_0$

Minima due to Single Slit Fringe Pattern ⇒

nth minima is obtained at

$$\theta = \sin^{-1}\left(\frac{n\lambda}{b}\right) \quad (n = 1, 2, 3, ---)$$

→ First minim is obtained at

$$\theta_1 = \frac{\lambda}{b}$$

[Considering v. small angles $\theta = \sin^{-1}\left(\frac{\lambda}{b}\right) \simeq \frac{\lambda}{b}$]

→ So, angular width of central bright fringe is

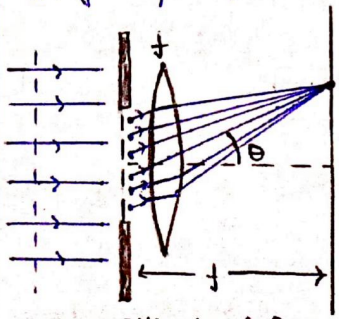
$$\phi = 2\theta_1 = \frac{2\lambda}{b}$$

Fringe Pattern of Diffraction due to Single Slit :

→ To perform diffraction practically, we need to take a thin convex lens.

→ Convex lens does not cause any path difference in parallel light beams.

→ Since parallel beams would interfere at focal plane, screen kept at 'f'.



Here,

$$I_P = I_0 \frac{\sin^2\beta}{\beta}$$

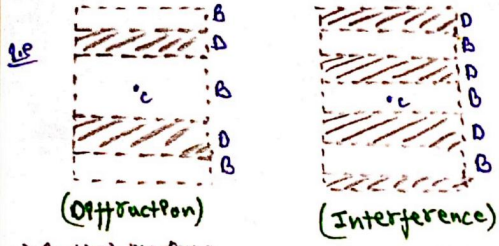
$$\beta = \frac{\pi b \sin\theta}{\lambda}$$

Angular width of minima ⇒

$$\sin\theta = n\lambda/b \quad (n \neq 0)$$

Wave-Optics 2

→ As we move far from the centre, width of bright fringes decreases.



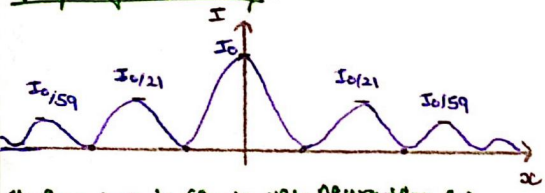
(Diffraction) (Interference)

→ Central maxima is largest.

→ Size of other max. ↓ses as their distance from centre ↑ses.

→ All fringes of same width

Graph of Intensity :→



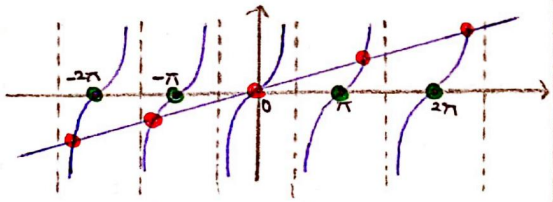
Maxima due to single slit Diffraction :→

For $I(\theta) = \frac{I_0 \sin^2\beta}{\beta}$ $\left(\text{where } \beta = \frac{\pi b \sin\theta}{\lambda}\right)$

maxima ies at the solution of eqn

$\boxed{\beta = \tan\beta}$

(Can be solve graphically)



● → Point of Maximas

● → Point of Minimas

Illumination pattern due to single slit Diffraction :

First Minima at $\theta_1 = \sin^{-1}(\lambda/b)$

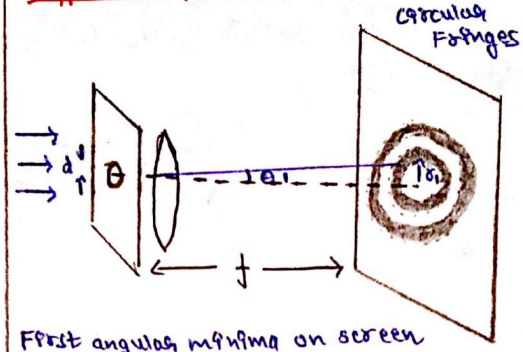
1) If $b \gg \lambda$ ⇒ slit size very large
⇒ $\theta_1 \simeq 0$ (No diffraction)

2) If $b > \lambda$ ⇒ comparable slit size
⇒ $\theta_1 \to$ Finite (Prominent diffraction)

3) If $b = \lambda$ ⇒ $\theta_1 = \pi/2$
⇒ No dark fring seen. ~~Whole screen has gradually fading bright fringe~~

4) If $b < \lambda$ ⇒ Same illumination on whole screen fading gradually on either side of central point.

Diffraction by a Circular Apperture :



First angular minima on screen obtained at

$\boxed{\theta_1 = \frac{1.22\lambda}{d}}$ (d → Diameter of apperture)

Radius of central Maxima (Airy Disc)

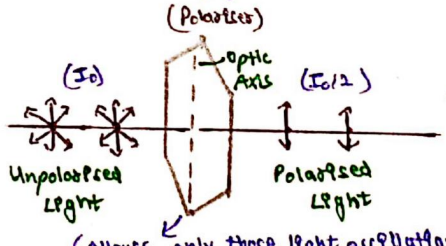
$\boxed{r_1 = f\theta_1 = \frac{1.22\lambda f}{d}}$

Polarization of Light :

→ Light is an EM wave with constantly oscillating EF & MF in all dirn.

→ EF is major contributor for us to see light.

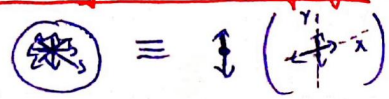
→ Thus while talking of polarization, we show only EF of light in all directions.



(Allows, only those light oscillations to pass, which are parall to its optic axis)

Polarisation : Restriction of oscillations of light (mainly $\vec{E}$) in one plane only.

Methods of Polarization of Light :

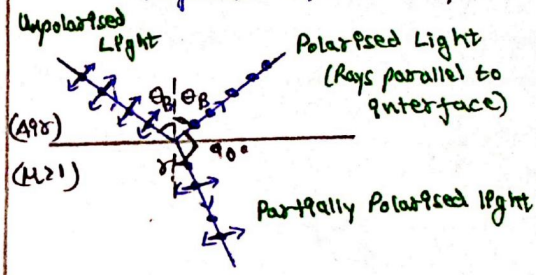


The unpolarised light can be resolved entirely into x & y components & represented as above.

1) Polarization by Reflection :→

→ When a light beam is incident on the interface of an optically denser med. at an angle (θ_B) such that the reflected and refracted ray are ⊥ to each other.

→ Then, Reflected Ray → Polarised
Refracted Ray ⇒ Partially Polarised



Brewster's Law ⇒

Gives angle of incidence at which reflected ray will be polarised

$\boxed{\tan\theta_B = \mu}$ (θ_B → Brewster's Angle)

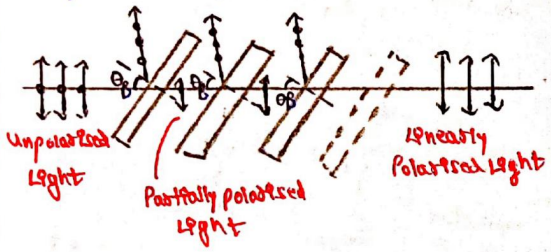
2) Polarization by Refraction :→

→ Here, we take a system of multiple parralel glass plates of equal thickness.

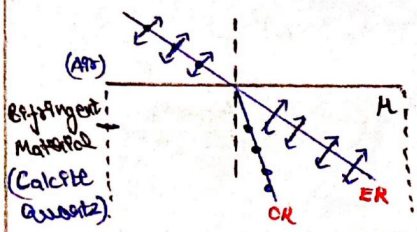
→ Light is incident on the plates at Brewster's Angle (θ_B).

→ Due to this, one of the ⊥ar component of light is refracted out multiple times.

→ And, after multiple reflections, polarised light in the dirn of original unpolarised light is obtained.



3) Polarization by Double Refraction :→



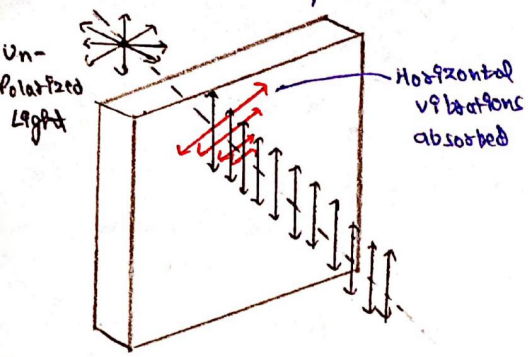
OR : Ordinary Ray (Follows law of Refraction, bends)
⇒ Polarises normal to the plane

ER : Extraordinary Ray (Not follows laws of refraction)
⇒ (Polarises in the plane)

4) Polarization by Dichroism :→

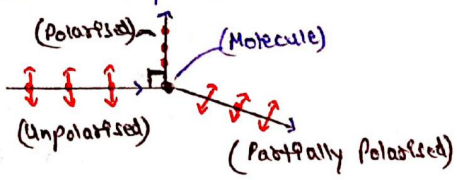
→ Dichroism is also called Horizontal vibration absorber.

→ It allows only vibrations ⊥ar to its axis, to pass throught it absorbs vibrations parallel to the axis.



5) Polarization by Scattering :→

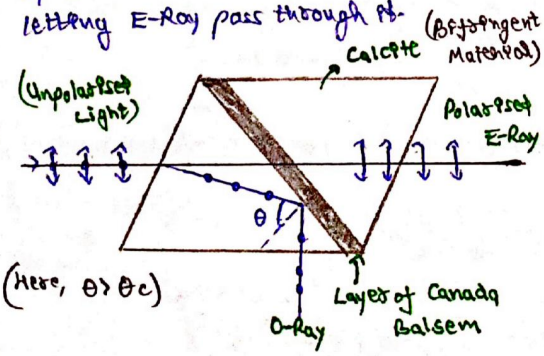
Light scattered at 90° is polarised ⊥ar to the plane.



Nicole Prism :

→ It is made by joining two trihedral prism of Calcite by a layer of cement called Canada Balsam.

→ Nicole prism behaves as a polariser by eliminatin O-Ray via TIR and letting E-Ray pass through it.



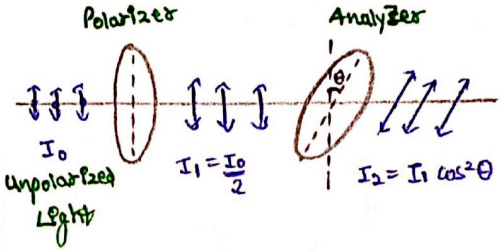
Malus Law :

Gives relation b/w intensities of polarised light.

→ If, a polarized light of intensity I_0 passed through an Analyser at θ, then intensity of new polarized light is given by

$$\boxed{I_p = I_0 \cos^2\theta}$$

NOTE :

(i) A polariser kept infront of an unpolarised light is called 'polariser' only

(ii) A polariser kept infront of polarized light is called an 'Analyser'